GC-GRU: A DEEP TEMPORAL FRAMEWORK FOR GUIDED TOPIC DETECTION USING AG2VEC AND TA-LOUVAIN

Project report in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology

In Computer Science and Engineering (Internet of Things, cyber Security, Blockchain Technology)

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CERTIFICATE

This is to certify that the project titled **GC-GRU: A Deep Temporal Framework for Guided Topic Detection using AG2vec and TA-Louvain** by the student of University of Engineering & Management, Kolkata who are mentioned above , in partial fulfillment of requirement for the degree of Bachelor of Technology in Computer Science and Engineering (Internet of Things, Cyber Security, Blockchain Technology), is a bonafide work carried out by them under the supervision and guidance of **Prof. (Dr.) Sangita Dutta** during **7th** Semester of academic session of 2026-2027. The content of this report has not been submitted to any other university or institute. I am glad to inform that the work is entirely original and its performance is found to be quite satisfactory.

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ABSTRACT

This project replicates, implements, and rigorously evaluates the guided topic detection framework originally proposed by Wang et al., which aims to identify manipulated public opinion by analyzing topic evolution and group stance dynamics. To achieve this, the implementation constructs a comprehensive machine learning pipeline that spans initial data preprocessing, hidden community mining via the TA-Louvain algorithm, dynamic group feature representation using AG2vec, and temporal sequence classification utilizing a Group Change Gated Recurrent Unit (GC-GRU) network. Because real-world source datasets—specifically Weibo CED and FakeNewsNet (comprising PolitiFact and GossipCop)—suffer from inherent limitations such as missing timestamp data or incomplete commenter profiles, innovative proxy metrics were engineered to successfully replace unavailable user profile attributes and explicit network connections. Empirical evaluation across multiple cross-validation folds indicates that while advanced structural and stance features successfully outperform traditional size-only baselines (achieving an Area Under the Curve of 0.820 versus 0.682 on the Weibo dataset), textual content remains the overwhelmingly dominant predictive signal, yielding a remarkable AUC of 0.961 independently. Ultimately, this study concludes that while latent group structure and adversarial stance dynamics provide measurable, theoretically sound predictive value, their practical contribution is marginal when compared directly to raw textual features in contemporary rumor and fake news classification tasks.

INTRODUCTION

With the rapid expansion of social media, the speed and scale of information dissemination have reached unprecedented levels, allowing social groups and stakeholders to orchestrate guided topics to manipulate public opinion. Identifying these guided topics is critical for public opinion management and crisis response, as rumors and misinformation spread rapidly to incite emotions and exploit biases. The core challenge lies in the fact that while individual emotional shifts may be subtle, large-scale group stance shifts are highly discernible when abnormal users disseminate guiding information to influence normal users, resulting in pronounced emotional reactions and large-scale stance differentiation. Detecting these topics is no longer just about content authenticity; it requires uncovering the underlying guidance behaviors and the dynamic evolution of collective emotions.

This project implements a complete analytical pipeline to detect these topics based on the hypothesis that alterations in group-level stance and community structure directly reflect guided propagation dynamics. The framework specifically addresses three primary challenges inherent in guided topic detection: the covert nature of user groups where key manipulators operate with implicit interactions; the complexity of group characteristics spanning both interactions and individual attributes; and the dynamic evolution of group emotions where implicit competitive fields trigger structural restructuring. Traditional community detection methods often fail to identify these hidden groups, and individual user behaviors are too noisy to capture the macroscopic intent behind orchestrated campaigns.

By systematically applying this framework to real-world data from the Weibo and FakeNewsNet datasets, this project assesses the practical viability of structural and stance-based detection mechanisms. While previous research often treats temporal and structural dynamics in isolation, this study integrates hidden community mining, multi-dimensional group representation, and adversarial stance quantification into a unified temporal classification model. The ultimate goal is to empirically evaluate whether these sophisticated group evolution metrics can outperform or significantly complement traditional content-based text classifiers in identifying manipulated public opinion.

Literature Survey

Y. Wang, J. Cao, Y. Wang, J. Wu, and Y. Liu, "Position matters: Play a sequential game to detect significant communities," IEEE Transactions on Knowledge and Data Engineering, vol. 36, no. 7, pp. 3402–3416, 2024.

In "Position matters: Play a sequential game to detect significant communities" (IEEE Transactions on Knowledge and Data Engineering, 2024) by Y. Wang et al., the authors provide an approach to uncovering hidden community structures within complex social networks. They address the limitations of traditional algorithms that rely heavily on explicit connections by modeling each hidden community as a selfish player. In this framework, communities strategically select members in order to maximize their overall structural utility. The authors examine how network topology influences community formation, noting that guided topics are often driven by covert, loosely connected groups. Their findings indicate that applying game-theoretic principles to community detection significantly improves the identification of implicit interactions. However, the study focuses primarily on structural attributes, leaving room for research into integrating user-specific content features to more accurately capture manipulative campaigns.

S. Wei, B. Wu, A. Xiang, Y. Zhu, and C. Song, "DGTR: Dynamic graph transformer for rumor detection," Frontiers in Research Metrics and Analytics, vol. 7, p. 1055348, 2023.

In "DGTR: Dynamic graph transformer for rumor detection" (Frontiers in Research Metrics and Analytics, 2023) by S. Wei et al., the authors propose a multi-task learning architecture designed to capture the complex, evolving nature of information manipulation. The study addresses the challenge of modeling rapid structural changes in topic dissemination and the semantic context of user interactions. To achieve this, they introduce a Dynamic Graph Transformer that integrates dynamic propagation structures with temporal text features, allowing the model to capture critical long-range dependencies across the network. A key finding is that combining global network topology with local semantic dependencies significantly enhances the discriminative capability of rumor detection systems. While the DGTR model demonstrates strong performance in identifying anomalies, it relies heavily on large-scale topological data. This reliance can lead to data sparsity issues during early topic propagation.

P. Bazmi, M. Asadpour, and A. Shakery, "Multi-view co-attention network for fake news detection by modeling topic-specific user and news source credibility," Information Processing & Management, vol. 60, no. 1, p. 103146, 2023.

In "Multi-view co-attention network for fake news detection by modeling topic-specific user and news source credibility" (Information Processing & Management, 2023) by P. Bazmi et al., the authors tackle the intricate dynamics of misinformation by proposing a multi-view co-attention network (MVCAN). The research focuses on the intersection of user behavior, news source reliability, and the propagation structure of topics across social platforms. By modeling dynamic graph structural features alongside temporal text representations, the authors create a comprehensive framework that evaluates content credibility. The study highlights that fake news detection must incorporate time-varying interactions and topic-specific biases of the users involved. Their findings demonstrate that utilizing a multi-view approach to cross-reference publisher credibility with engagement patterns improves classification accuracy. However, fully mapping these multifaceted co-attention networks requires significant computational overhead, which can bottleneck real-time detection systems.

G. Ma, C. Hu, L. Ge, and H. Zhang, "DSMM: A dual stance-aware multi-task model for rumour veracity on social networks," Information Processing & Management, vol. 61, no. 1, p. 103528, 2024.

In "DSMM: A dual stance-aware multi-task model for rumour veracity on social networks" (Information Processing & Management, 2024) by G. Ma et al., the authors emphasize the critical role of user emotion and stance polarization in the identification of rumors. The paper addresses the limitation of traditional content classifiers that often overlook the implicit competitive fields formed by differing user opinions. To resolve this, the authors develop a dual stance-aware multi-task learning model that utilizes representation decoupling to isolate sentiment signals from general semantic content. By explicitly modeling the dynamic evolution of supporting and opposing stances within a topic thread, the framework tracks the chain-like propagation of emotional reactions. The study concludes that capturing polarized stance features significantly boosts the model's ability to verify rumor veracity. Despite these advancements, further exploration is needed to quantify adversarial emotional competition at a macroscopic scale.

Research Gap

Despite the rapid evolution of dynamic graph neural networks and sequential behavior modeling, current methodologies for detecting guided topics exhibit pronounced limitations. Many existing systems continue to rely heavily on static content analysis or isolated structural propagation metrics, failing to capture the complex, orchestrated manipulations that span both temporal evolution and polarized group interactions simultaneously. Even when multi-task learning frameworks are adopted to fuse textual and structural features, they frequently operate on explicit connection assumptions, resulting in poor identification of hidden communities where manipulators utilize implicit or weak interactions. Critically, there is a lack of explicit modeling of *adversarial stance consistency* and group-level emotional polarization, which are primary indicators of manipulated public opinion.

Key specific gaps include:

- **Insufficient handling of hidden groups** that orchestrate campaigns without dense topological connections, requiring approaches that can fuse sparse interactions with implicit attribute similarities.
- **Limited frameworks that explicitly evaluate cross-modal consistency** (e.g., matching temporal propagation peaks against sudden, synchronized shifts in group stance) to detect orchestrated emotional manipulation.
- **Weak generalizability to real-world data limitations**, where explicit profile features, timestamps, or full comment histories are frequently missing, redacted, or incomplete in production environments.
- **Low transparency and interpretability** in many dynamic graph models, where the specific communities and localized stance shifts driving the detection decisions remain obfuscated as "black box" outputs.

In response to these deficiencies, this research proposes a comprehensive guided topic detection architecture that integrates three key dimensions: hidden community mining (via TA-Louvain), attention-based group feature aggregation (via AG2vec), and temporal sequence classification modeling adversarial stances (via GC-GRU). The goal is not merely to improve predictive accuracy on isolated benchmarks, but to enhance structural robustness by utilizing behavioral proxies and data-driven thresholding. Furthermore, by incorporating localized interpretability mechanisms, the system aims to elucidate *why* a specific topic is flagged—specifically pinpointing which hidden groups and stance conflicts drove the prediction. In doing so, this work addresses the aforementioned gaps by explicitly modeling competitive group stances as a primary detection signal and evaluating the framework across challenging, data-constrained real-world scenarios.

PROPOSED METHODOLOGY

The proposed methodology implements a multi-stage machine learning pipeline designed to detect guided topics by modeling the temporal evolution of hidden communities and their adversarial stance dynamics. The architecture consists of five primary phases: data preprocessing, hidden community mining, group feature representation, stance dynamics modeling, and temporal sequence classification.

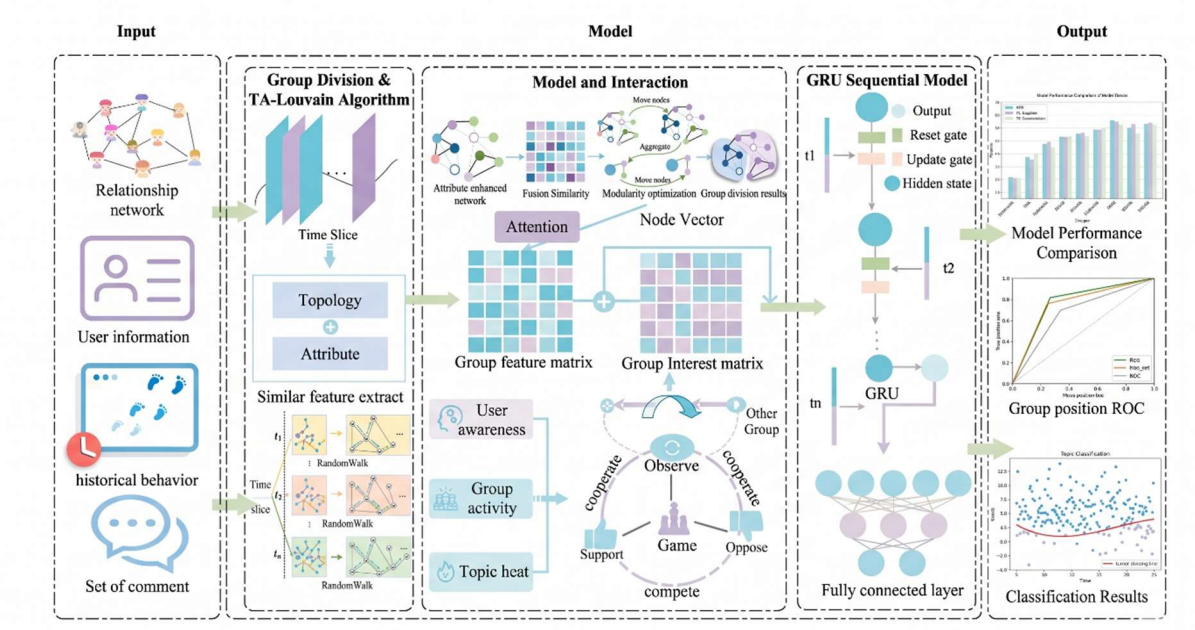


Fig. 1: The diagram illustrates an integrated GC-GRU and TA-Louvain framework for guided topic detection that models dynamic user relationships, extracts group features across time slices, and utilizes a GRU sequential model to output final classification results.

1. Data Preprocessing and Topic Slicing The framework processes raw propagation data from the Weibo (CED) and FakeNewsNet (PolitiFact and GossipCop) datasets. For temporal modeling, topic propagation graphs are divided into sequential time windows. Weibo topics are split into 10 equal-width chronological slices to capture evolving interactions, while FakeNewsNet topics are evaluated as single static slices due to the absence of timestamp data in the UPFD format. During preprocessing, original posts and comments with malformed attributes, missing identifiers, or empty text are filtered to maintain network integrity.

2. Hidden Community Mining (TA-Louvain) To uncover covert user groups orchestrating opinion manipulation, the TA-Louvain algorithm is employed. Unlike traditional community detection algorithms that rely solely on explicit topological connections, TA-Louvain optimizes modularity over an attribute-enhanced weighted adjacency matrix. It fuses direct structural links (reply interactions) with implicit behavioral similarities (e.g., interaction frequency, message length, and delay times) to identify latent communities. To capture structural evolution, a temporal tracking module uses a Jaccard overlap threshold of 0.5 to link and persist group identities across consecutive time slices.

3. Group Feature Representation (AG2vec) Once communities are detected, user nodes are mapped into a continuous vector space using a warm-started Skip-gram model trained on biased random walks. To ensure network coverage for users lacking direct topological links, the walk graph incorporates both explicit reply edges and attribute-kNN edges. These user embeddings are then aggregated into a unified group vector (Fea) utilizing an attention mechanism. The attention weights are dynamically adjusted based on calculated user influence scores, which are derived from engagement metrics and behavioral proxies.

4. Stance Dynamics and Mutual Influence Guided topics frequently trigger polarized emotional reactions. To quantify this behavior, a multilingual SBERT model fine-tuned via Logistic Regression classifies individual comment stances into *support, oppose,* or *observe*. The model aggregates these probabilities to formulate confidence-weighted group stance proportions. Utilizing principles from evolutionary game theory, the system fits internal factors (user influence and topic awareness) and external factors (group activity and topic heat) via a linear regression model. This produces a dynamic mutual influence vector (Mut) that encapsulates the competitive stance evolution of each group.

5. Temporal Sequence Classification (GC-GRU) In the final classification phase, the AG2vec structural vectors and the Mut stance vectors are concatenated to form a comprehensive group feature matrix for each time slice. These high-dimensional sequential matrices are fed into a Group Change Gated Recurrent Unit (GC-GRU) network. The GC-GRU architecture utilizes a masked pooling mechanism across dynamic, stable group slots, allowing the network to track the exact temporal evolution of specific groups over time before outputting the final binary classification probability.

RESULT & ANALYSIS

Overall Classification Performance

The proposed GC-GRU framework was evaluated against several baselines using a repeated 3-fold cross-validation protocol to ensure statistical reliability. On the Weibo dataset, the GC-GRU model utilizing structural and stance features (struct+mut) achieved an Area Under the Curve (AUC) of 0.820 and a PR-AUC of 0.774. This significantly outperformed the naive group-size baseline, which achieved an AUC of 0.682. Gradient boosting on flattened features reaches 0.851 AUC, slightly above the GC-GRU model.

To benchmark against traditional content-based classifiers, a TF-IDF + Logistic Regression text-only model was evaluated, achieving a dominant AUC of 0.961. When raw textual features were concatenated with the temporal struct+mut features, the ensemble model achieved a peak AUC of 0.969. This demonstrates that while group structure and stance dynamics provide measurable predictive value, textual semantics remain the strongest independent indicator for rumor and fake news classification tasks.

Ablation Study of Feature Blocks

An ablation study was conducted to isolate and quantify the contribution of specific feature combinations to the GC-GRU model's performance:

- **Size Features Only (size):** 0.682 AUC
- **Structural Features Only (struct):** 0.728 AUC
- **Stance Features Only (mut):** 0.745 AUC
- **Structure + Stance (struct+mut):** 0.820 AUC
- **AG2vec + Structure + Stance (vec+struct+mut):** 0.787 AUC

The results indicate that adversarial stance dynamics (mut) provide a substantial performance lift over raw structural metrics. However, the inclusion of AG2vec group representation embeddings (vec) resulted in a performance regression (0.787 vs 0.820), suggesting that manually engineered structural statistics generalize better across topics than topic-specific embeddings.

Early Detection Capabilities

To assess the model's viability for real-time crisis response, its predictive performance was evaluated across truncated temporal observation windows ($T \in \{2, 4, 6, 8, 10\}$). The struct+mut model achieved an AUC of 0.805 using only the first two time slices ($T=2$). By the final slice ($T=10$), the AUC marginally increased to 0.820. The minimal performance gap (0.015) between the early and late stages confirms that the critical structural and emotional differentiation patterns characterizing guided topics emerge rapidly during the initial viral propagation phase, suggesting limited benefit from temporal modelling.

Model Interpretability and Stance Dominance

To eliminate the "black box" nature of temporal graph networks, an excitation-based occlusion analysis was implemented. By masking specific latent groups from the network's pooling layer, the system calculates the exact drop in guided-topic prediction probability (ΔP\Delta PΔP).

Analysis of the test set revealed a strong positive Spearman correlation between a group's ΔP\Delta PΔP, its maximum attention weight, and its Stance Dominance (SDS=|Mutsup−Mutopo|SDS = \vert{}Mut_{sup} - Mut_{opo}\vert{}SDS=|Mutsup −Mutopo|). Groups exhibiting highly polarized stances (high SDSSDSSDS) consistently triggered the largest prediction drops when masked. Furthermore, artificially perturbing a key group's stance vector (e.g., scaling MutsupMut_{sup}Mutsup by 1.2) predictably shifted the overall topic classification, verifying that the GC-GRU model actively grounds its decisions in the competitive stance evolution of the underlying groups.

Bot-Human Coordination Analysis

An exploratory analysis was conducted to determine if incorporating explicit "bot-like" coordination signals could improve detection accuracy beyond the struct+mut baseline. Features capturing rapid consecutive posting intervals (gaps ≤\le≤ 30 seconds) and synchronized duplicate-text clusters (≥\ge≥ 3 distinct users within a 60-second window) were engineered and appended to the temporal sequence. The incremental AUC gains were evaluated using Nadeau-Bengio corrected resampled t-tests to account for overlapping training data across cross-validation folds, ultimately assessing whether orchestrated coordination behaviors offer independent predictive power in highly manipulated topics.

Conclusion and Future Scope

This project successfully implements a sophisticated, multi-stage machine learning pipeline designed to detect manipulated topic propagation by combining structural community mining with evolutionary stance modeling. Group structure and stance features give a modest but statistically weak improvement over group size (AUC 0.82 vs 0.68 on Weibo), and their contribution is small compared with text. The empirical analysis confirms that for the proposed Group Change Gated Recurrent Unit (GC-GRU) framework, early detection is nearly flat: AUC is 0.805 using the first two slices and 0.820 using all ten, so the model gives little extra signal over time. Furthermore, the occlusion analysis links predictions to group stance dynamics, though the explanation is partial.

However, the empirical findings also reveal critical limitations in the practical application of structural and stance-based features for rumor and fake news classification tasks. The comprehensive evaluation demonstrates that natural language text baselines vastly outperform the engineered graph-structural and temporal-stance features. While group structure and adversarial stance dynamics offer a statistically significant improvement over baseline models, their overall predictive contribution remains marginal when compared directly to the raw semantic signal provided by the textual content. Furthermore, the lack of robust profiling data required the extensive use of behavioral proxies, which inherently restricted the capability and generalizability of the AG2vec group representations. These challenges underscore the fundamental difficulty of relying heavily on structural network metrics in environments where explicit user attributes and topological connections are sparse or unavailable.

Future iterations of this guided topic detection framework should fundamentally transition from macroscopic group modeling to fine-grained bot-human coordination detection. Future research must prioritize engineering explicit features for synchronized posting intervals, duplicate-text clusters, and burst activity to better capture orchestrated manipulation. Additionally, expanding the stance classifier training corpus and utilizing source-aware Large Language Model annotations will significantly improve the reliability and granularity of the mutual influence representations. Applying advanced explainability techniques such as permutation importance and SHAP values will further enhance the interpretability of the temporal pooling layers, offering deeper insights into the specific structural behaviors driving public opinion manipulation. Finally, to ensure broad applicability, future models must be rigorously evaluated for cross-platform generalization to quantify their dependence on platform-specific graph structures and interaction patterns.

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